

Bode Plot & Simple Filter Design

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Abstract:

A first-order low-pass filter with a voltage gain of 10 and a cutoff frequency of 3 kHz was designed and verified in LTspice using an LT1630 operational amplifier in a non-inverting configuration. Simulations included the frequency response, step response, and sinusoidal responses at 30 Hz, 3 kHz, and 300 kHz. The LT1368 op-amp was initially used but replaced due to signal attenuation. The LTspice results closely matched the theoretical transfer function, with minor deviations due to op-amp limitations.

Design:

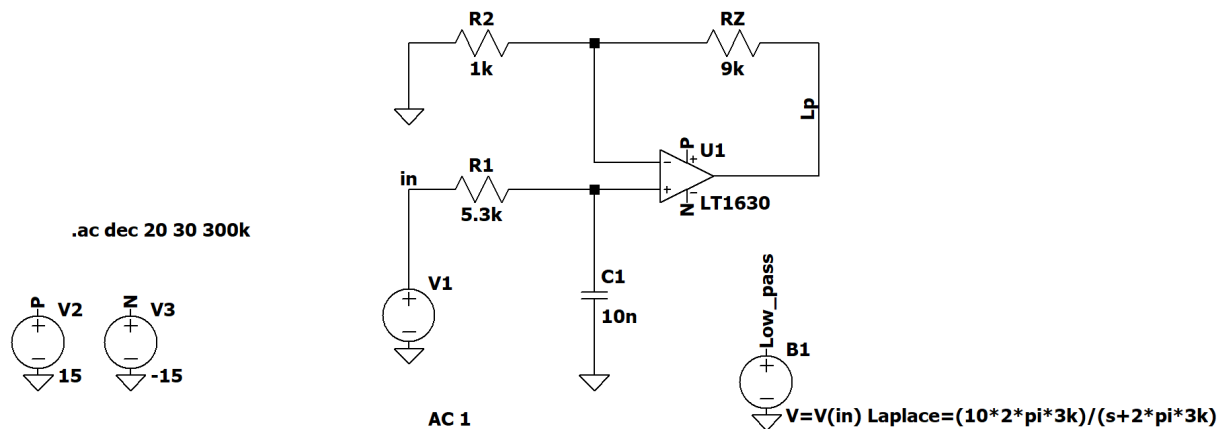


Figure 1: A first-order low-pass filter with a gain of 10 and a cutoff frequency of 3 kHz

$$H(s) = \frac{V_{out}}{V_{in}} = \frac{A_v}{1 + sRC} \quad (\text{eq. 1})$$

Where:

$$A_v = 1 + \frac{R_f}{R_2} \quad (\text{eq. 2})$$

$$H(s) = \frac{V_{out}}{V_{in}} = \text{gain} \times \frac{\omega_0}{\omega_0 + s} \quad (\text{eq. 3})$$

$$\omega_0 = 2\pi f_0$$

$$f_0 = \frac{1}{2\pi \times R_1 C_1}$$

$$\omega_0 = 2\pi \times \frac{1}{2\pi \times R_1 C_1} = \frac{1}{R_1 C_1}$$

Gain = 10

$$H(s) = 10 \times \frac{\frac{1}{R_1 C_1}}{\frac{1}{R_1 C_1} + s} \quad (\text{eq. 4})$$

Cutoff frequency f_c is defined as:

$$f_c = \frac{1}{2\pi RC} \quad (\text{eq. 5})$$

For the design, the target parameters were $A_v = 10$ and $f_c = 3 \text{ kHz}$. The values were chosen by rearranging equation (5). From this, $RC = 5.03 \times 10^{-5} \text{ s}$.

A standard capacitor value $C_1 = 10 \text{ nF}$

$$R_1 = \frac{1}{2\pi \times f_c \times C_1} = \frac{1}{2\pi(3 \text{ kHz})(10 \text{ nF})} \simeq 5.3 \text{ k}\Omega \quad (\text{eq. 6})$$

A standard $5.3 \text{ k}\Omega$ resistor was selected.

To achieve the desired gain of 10, the ratio of the feedback resistor from equation (2).

$$A_v = 1 + \frac{R_f}{R_2} \quad (4)$$

Choosing $R_2 = 10 \text{ k}\Omega$ && $R_f = 90 \text{ k}\Omega$

Using the transfer function from equation (4)

$$Y(s) = \frac{1}{s} \times 10 \times \frac{\frac{1}{R_1 C_1}}{\frac{1}{R_1 C_1} + s}$$

Now taking the inverse Laplace

$$y(t) = L^{-1} \left\{ \frac{1}{s} \times 10 \times \frac{\frac{1}{R_1 C_1}}{\frac{1}{R_1 C_1} + s} \right\} = 10(1 - e^{-\frac{1}{R_1 C_1} t})u(t)$$

Where again, $R_1 \times C_1 = 5.03 \times 10^{-5} s$

The LT1368 op-amp was tested initially, but the amplitude response exhibited attenuation due to internal compensation capacitance. The LT1630, with a higher gain-bandwidth product (GBW), was substituted and provided accurate performance near the designed cutoff.

Verification:

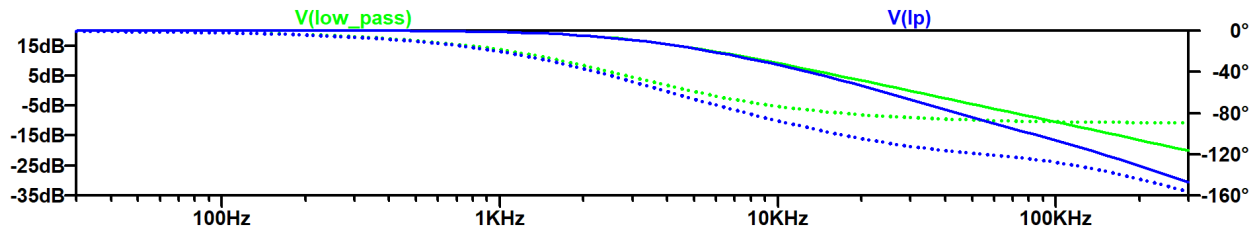


Figure 2. AC Analysis (Bode Plot) using LT1368

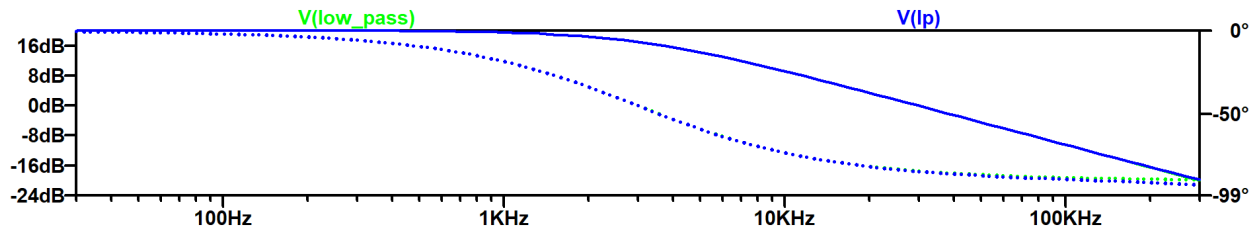


Figure 3. AC Analysis (Bode Plot) using LT1630

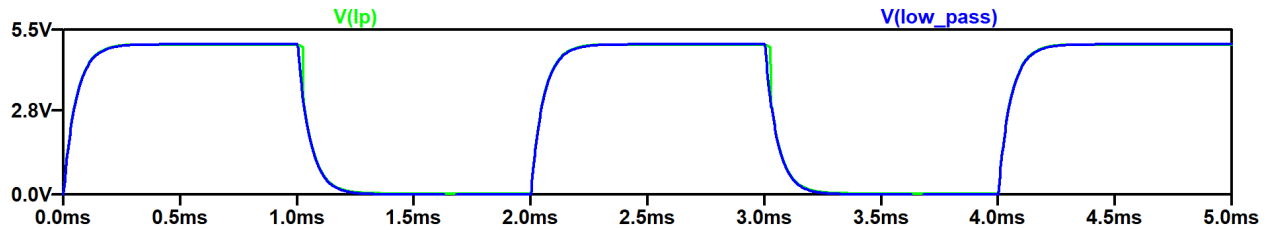


Figure 4. Transient Analysis (Step Response) using LT1630, where $\rightarrow$.tran 0 5m 0 10u
& $\rightarrow$ PULSE(0 5.5 0 1u 1u 1m 2m)

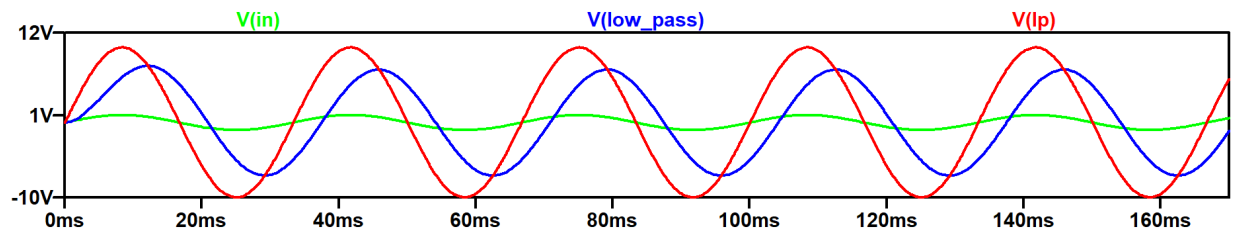


Figure 5. Sine Response using LT1630 of 30 Hz, where $\rightarrow$ Sine(0 1 30Hz)
& $\rightarrow$.tran 0 170m

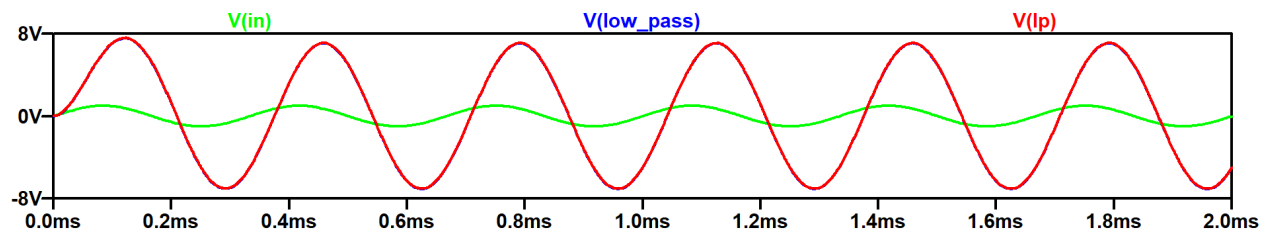


Figure 6. Sine Response using LT1630 of 3 kHz, where $\rightarrow$ Sine(0 1 3 kHz)
& $\rightarrow$.tran 0 2m

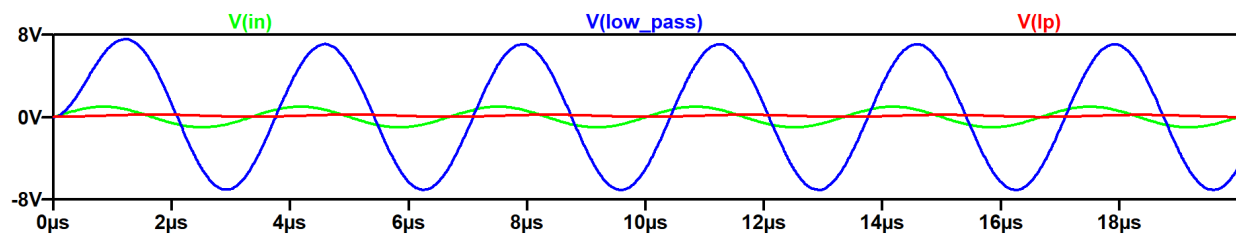


Figure 6. Sine Response using LT1630 of 300 kHz, where $\rightarrow$ Sine(0 1 300 kHz)
& $\rightarrow$.tran 0 20u

Discussion/Summary:

The simulation results matched the theoretical low-pass filter response closely. The Bode plot indicated a -20 dB/decade roll-off beyond the 3 kHz cutoff, and the passband gain was measured at approximately 20 dB, consistent with the designed gain of 10.

Initial trials with the LT1368 op-amp showed reduced output amplitude due to its internal compensation capacitance affecting high-frequency performance. The LT1630, with a higher GBW, provided the expected frequency response. Minor discrepancies between theoretical and simulated results can be attributed to the finite bandwidth of the op-amp, input offset voltages, and parasitic capacitances in the model.

A first-order low-pass filter with a gain of 10 and a cutoff frequency of 3 kHz was successfully designed and verified using LTspice. The filter exhibited the expected frequency-dependent attenuation and phase characteristics. Replacing the LT1368 with an LT1630 resolved the attenuation issue and produced results consistent with theoretical predictions. The lab demonstrated the relationship between op-amp parameters, filter performance, and real-world non-idealities.
